1. System Development Methodology

For the University Transport Route Tracking System, we select the Agile (Scrum) Methodology combined with Iterative Development.

- • Continuous Feedback: Allows regular testing and improvements for real-time GPS tracking and map integrations.
- • Flexibility: Adapts easily to changing requirements from university administration and student feedback.
- • Modular Development: The system is naturally divided into distinct modules that can be built and tested incrementally (Student Mobile App, Driver App, Admin Web Dashboard).

2. Time Estimation&Work Breakdown Structure

Total Estimated Project Duration: 12 Weeks (3 Months).

Phase / Task	Description	Duration	Dependencies
Phase 1: Planning&Feasibility	System Request, Cost-Benefit Analysis, Scope Definition	Weeks 1 - 2	None
Phase 2: Requirement Analysis	Use Cases, Database Schema, UI/UX Wireframes	Weeks 3 - 4	Phase 1
Phase 3: Core Development	Backend API, Database Setup, Admin&Apps	Weeks 5 - 8	Phase 2
Phase 4: Integration&Testing	GPS Hardware&Google Maps API sync, Bug Fixing	Weeks 9 - 10	Phase 3
Phase 5: Deployment	Cloud Server Deployment, Driver Training, Delivery	Weeks 11 - 12	Phase 4

3. Case Study&Use Case Scenarios

- • Student: Uses the mobile app to view bus locations and ETAs.
- • Bus Driver: Uses the driver app to start/end routes and report delays.
- • Transport Administrator: Manages routes, schedules, and user accounts via the web dashboard.

Use Case 1: Real-Time Bus Tracking (Student)
• Trigger: Student opens the mobile app and selects their campus route.
• Main Flow: System fetches live GPS coordinates, calculates ETA, and displays the moving bus on the map.

Use Case 2: Route&Schedule Management (Administrator)
• Trigger: Admin logs into the Web Dashboard to update morning schedules.
• Main Flow: Admin modifies route waypoints or assigns a backup bus, broadcasting instant updates.

4. PERT Chart Analysis

Expected Time formula: TE = (O + 4M + P) / 6
• Task A (Requirement Analysis): TE = 2.0 Weeks
• Task B (UI/UX&Design): TE = 2.0 Weeks
• Task C (Core Development): TE = 4.0 Weeks
• Task D (GPS&Maps Integration): TE = 2.0 Weeks
• Task E (Testing&Deployment): TE = 2.0 Weeks

Critical Path: Start ->Analysis (2w) ->Design (2w) ->Core Development (4w) ->Integration (2w) ->Testing&Deployment (2w) = 12 Weeks Total.

5. Gantt Chart Schedule

Project Phase	Duration	Timeline (Weeks 1 - 12)
1. Planning&Feasibility	2 Weeks	[Week 1 - 2]
2. Analysis&Architecture	2 Weeks	[Week 3 - 4]
3. Core App Development	4 Weeks	[Week 5 - 8]
4. GPS&Maps Integration	2 Weeks	[Week 9 - 10]
5. Testing&Deployment	2 Weeks	[Week 11 - 12]